

Adv. Set Theory and Cardinality

Zermelo Set Theory

The only non-logical primitive is the set membership relation $\in$.

All other set-theoretic notions defined ultimately in terms of $\in$. E.g.,

$$x \subseteq y =_{df} \forall z(z \in x \rightarrow z \in y).$$

(Extensionality) Sets having the same elements are identical.

(Separation Schema) For any set A and any property $P()$ involving ultimately only $\in$, there exists the set $\{x \in A \mid P(x)\}$.

N.B. A property alone defines a **class**. Some classes are “too big” to count as sets. E.g., Take the class $C = \{x \mid x \notin x\}$.

If C is set, we are led to a contradiction: $C \in C$ iff $C \notin C$. So, C is said to be a **proper** class.

How does that help? Although sets are allowed to be elements of other sets (and of classes), classes are not allowed to be elements of classes (and certainly not sets). And the variable in expression such as x in $\{x \mid x \notin x\}$ ranges over only sets.

Proposition. There is no set of all sets.

Proof. Suppose there is, call it U . Then $D = \{x \in U \mid x \notin x\}$ is a set and we arrive at the contradiction $D \in D$ iff $D \notin D$.

Further Axioms of $\mathbf{Z}$

(Unordered Pairs) For all x and y , $\{x, y\}$ is a set. (An *ordered* pair (x, y) is defined to be $\{\{x\}, \{x, y\}\}$.)

(Union) For any set A , $\bigcup A$ is a set, where

$$\bigcup A =_{df} \{x \mid \exists y \in A \text{ s.t. } x \in y\}.$$

Def. $S(x) = x \cup \{x\}$.

Def. $0 = \emptyset$, $1 = S(0)$, $2 = S(1)$, etc.

In general, $n + 1 = \{0, \dots, n\}$.

(Infinity) There exists a set A s.t. $\emptyset \in A$ and for all x , if $x \in A$, then $S(x) \in A$.

(Power Set). For any set A , $\mathcal{P}(A) =_{df} \{B \mid B \subseteq A\}$ is a set.

(Foundation/Regularity) Every non-empty set x has a member y s.t. $x \cap y = \emptyset$.

$\mathbf{Z}$ as a Foundation for All of Mathematics

We've seen how to construct $\mathbb{N}$ in $\mathbf{Z}$. Given $\mathbb{N}$, there are standard ways to construct

- ▶ $\mathbb{Z}$ from $\mathbb{N}$,
- ▶ $\mathbb{Q}$ from $\mathbb{Z}$,
- ▶ $\mathbb{R}$ from $\mathbb{Q}$, and
- ▶ $\mathbb{C}$ from $\mathbb{R}$.

The Concept of Cardinality

The idea of the **cardinality** of a set A amounts to being able to answer the question: how many members does A have?

The answer presupposes that there are **cardinal numbers**. For finite sets, this is not particularly problematic. We can use natural numbers. For larger sets it takes some finesse.

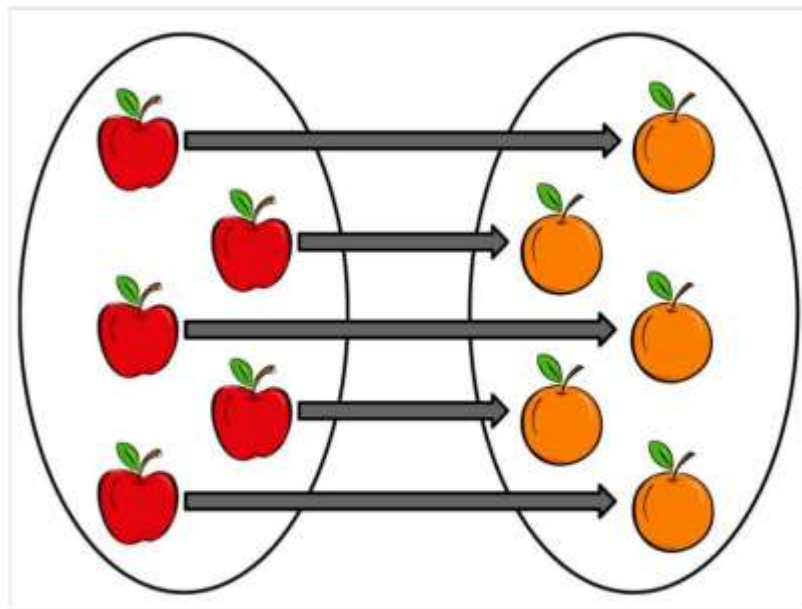
The answer also presupposes that we can make **cardinality comparisons**: do two sets have the same number of members or does one set have more than the other?

Cardinality comparisons can be defined even in the absence of cardinal numbers. They are a matter of whether certain **mappings** exist from one set to another.

Cardinality

In mathematics, **cardinality** is an inherent property of sets, roughly meaning the number of individual objects they contain, which may be infinite. The concept is understood through one-to-one correspondences between sets. That is, if their objects can be paired such that each object has a pair, and no object is paired more than once.

Two sets are said to be **equinumerous** or *have the same cardinality* if there exists a one-to-one correspondence between them. Otherwise, one is said to be *strictly larger* or *strictly smaller* than the other. Using this concept, it is possible to show there are different sizes of infinity.



A one-to-one correspondence between a set of apples and a set of oranges shows they have the same cardinality.

Functions and Their Inverses

Mappings involve functions. So, recall that a **function** f is a relation such that for every $x \in \text{dom } f$ there exists a unique y s.t. $(x, y) \in f$ and this unique y is denoted ' $f(x)$ '.

Every function f has an inverse f^{-1} , viz.

$$f^{-1} = \{(y, x) \mid (x, y) \in f\},$$

which, although a relation to be sure, need not itself be a function.

Def 1.4.1. A function f is said to be **one-to-one (1-1)** iff f^{-1} is also a function, or, equivalently, for all $x_1, x_2 \in \text{dom } f$,

$$\text{if } f(x_1) = f(x_2), \text{ then } x_1 = x_2.$$

The notion of a 1-1 function is crucial for making cardinality comparisons.

Mappings

Def 1.4.2. f **maps** A to B iff (i) f is a function s.t. (ii) $A = \text{dom } f$ and (iii) $\text{ran } f \subseteq B$.

B is called the **codomain** of the mapping.

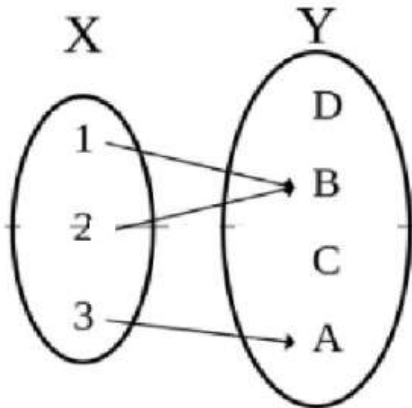
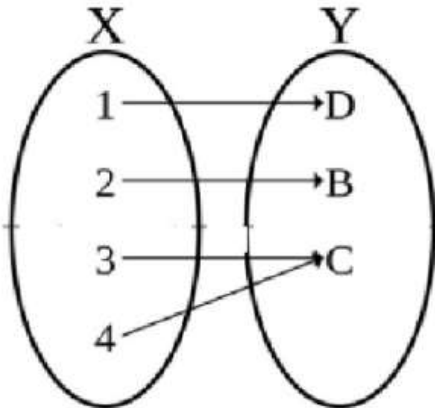
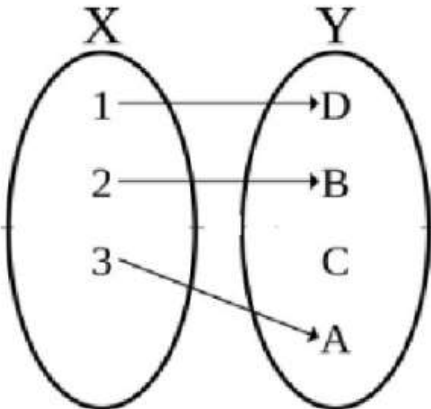
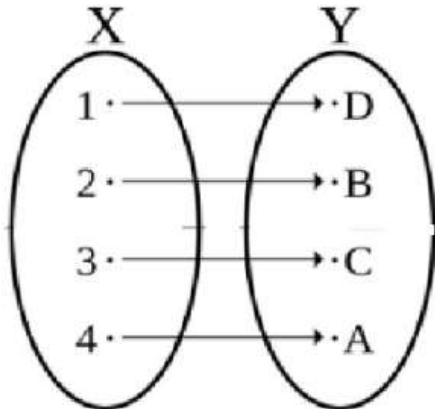
The notation ' $f : A \rightarrow B$ ' means either

- ' f maps A to B ', or, more commonly,
- 'the mapping f from A to B '.

Def 1.4.3.: If f is 1-1, then $f : A \rightarrow B$ is said to be an **injection**, or, simply, **1-1**.

Def 1.4.4. If $\text{ran } f = B$, then $f : A \rightarrow B$ is said to be a **surjection**, or, simply, **onto**.

If $f : A \rightarrow B$ is both injective and surjective, then $f : A \rightarrow B$ is said to be a **bijection**, or, a **1-1 correspondence** from A to B .

	not surjective	surjective
not injective	 general function	 surjective only
injective	 injective only	 bijective

Mappings (cont.)

Recall that, for relations R and S ,

$$S \circ R = \{(x, z) \mid \text{there exists a } y \text{ s.t. } (x, y) \in R \text{ and } (y, z) \in S\}.$$

Since functions are relations, we can use the same definition for the composition of functions, and hence the composition of mappings.

However, here is a more perspicuous definition:

Def 1.4.5. Suppose that $f : A \rightarrow B$ and $g : B \rightarrow C$. Then the **composition** of g with f is the mapping $g \circ f : A \rightarrow C$ s.t., for any $x \in A$, $(g \circ f)(x) = g(f(x))$.

Cardinality Comparisons

A cardinality comparison of one set with another involves the existence of certain mappings from the one set to the other.

Def 1.4.6. A is **equinumerous with** B , written $A \sim B$, iff there is a bijection $f : A \rightarrow B$.

Alternative terminology: A is **equipotent** with, **equipollent** with, or **similar** to B .

Example. $\{\text{Moe}, \text{Curly}, \text{Larry}\} \sim \{0, 1, 2\}$.

Sample bijection: $f(\text{Curly}) = 2$, $f(\text{Moe}) = 0$, $f(\text{Larry}) = 1$.

Cardinality Comparisons (cont.)

Def 1.4.7. *A is no more numerous than B*, written $A \preceq B$, iff there exists an injection $f : A \rightarrow B$.

An equivalent definition is that A is equinumerous with some subset of B .

Def 1.4.8. *A is less numerous than B*, written $A \prec B$, iff $A \preceq B$ but $B \not\preceq A$.

Example. $\{\text{Bonnie, Clyde}\} \preceq \{0, 1, 2\}$.

Sample injection: $f(\text{Bonnie}) = 0$, $f(\text{Clyde}) = 2$.

But $\{0, 1, 2\} \not\preceq \{\text{Bonnie, Clyde}\}$. No surjection is 1-1.

Hence $\{\text{Bonnie, Clyde}\} \prec \{0, 1, 2\}$.

Basic Properties of $\sim$

Lemma 1.4.1. For any sets A , B , and C ,

1. $A \sim A$.
2. If $A \sim B$, then $B \sim A$.
3. If $A \sim B$ and $B \sim C$, then $A \sim C$.

Proof Sketch:

For (1), use the identity map $id_A : A \rightarrow A$.

For (2), show that the inverse of a bijection is a bijection.

For (3), show that the composition of bijections is a bijection.

Basic Properties of $\sim$ (cont.)

Scholium.

So $\sim$ behaves formally like an equivalence relation.

Only $\sim$ is not really a relation on a set because (i) there is no set of all sets, and (ii) $\sim$ is not a SET of ordered pairs, and hence not a RELATION, but a PROPER CLASS of ordered pairs.

However, equinumerosity does partition V , the class of all sets, into equivalence classes, which also are not sets, but proper classes.

To illustrate, recall that $V = \{x \mid x \text{ is a set}\}$. If $\sim$ were a set, then $(\text{dom } \sim)$ would be a set, and hence so would V , since

$$(\text{dom } \sim) = \{x \mid x \in V\}.$$

Basic Properties of $\preceq$

Lemma 1.4.2. For any sets A , B , and C ,

1. $A \preceq A$
2. If $A \preceq B$ and $B \preceq C$, then $A \preceq C$.
3. If $A \preceq B$ and $B \preceq A$, then $A \sim B$.

Proof Comments: The proofs of (1) and (2) are rather immediate. For (1), the identity map $id_A : A \rightarrow A$ again suffices, and for (2), it suffices to show that the composition of injections is an injection.

Part (3), however, is highly non-trivial, and is known in the literature as the **Schröder-Bernstein Theorem**. One is given the existence of two injections, $f : A \rightarrow B$ and $g : B \rightarrow A$, neither of which can be assumed to be surjective, and then one must somehow construct a bijection $h : A \rightarrow B$.

Are All Sets Comparable in Cardinality?

Scholium: The preceding lemma suggests that $\preceq$ behaves like a partial ordering if one thinks of $\sim$ as replacing identity. So, without begging too much confusion, it says that V , the class of all sets, is partially ordered in cardinality (numerosity). This motivates the following question:

Query: Does it also follow that, for any sets A and B , $A \preceq B$ or $B \preceq A$, i.e., are any two sets comparable in cardinality (numerosity)?

It would seem strange if there were sets A and B without the same number of members, but neither of which has *more* members than the other.

Answer: No, not unless one assumes the axiom of choice (**AC**), which is equivalent to the proposition that all sets can be well-ordered.

(This, then, is strong motivation for adopting **AC**. That **AC** is non-trivial is evident from the following challenge: construct a well-ordering of $\mathcal{P}(\mathbb{N})$ with assuming **AC**.)

Finite, Infinite, and Uncountable Sets

Def 1.4.9. A set A is **finite** iff $A \sim n$ for some $n \in \mathbb{N}$, in which case it is said to have n members.

Otherwise, A is **infinite**.

Def 1.4.10. A is **countable** iff $A \preceq \mathbb{N}$.

Otherwise, A is **uncountable**.

N.B. Two important points deserve emphasis here:

1. One needs to distinguish between infinity in **number** and infinity in **magnitude** (for example, infinite distance, or infinite mass).
2. As for infinity in number, if there are uncountable sets, then some infinities in number are greater than others.

Countably Infinite Sets

Are there any uncountable sets?

To try to answer this, let us start going through our methods of constructing sets from other sets and see if, when applied to countably infinite sets, they do or do not yield a set that is countably infinite...

Lemma 1.4.3. The union of countable sets is countable.

The proof is left as an exercise.

Corollary. The set $\mathbb{Z}$ of integers is countable. [Defn. of $\mathbb{Z}$? Next slide.]

Proof. $\mathbb{Z}$ is the union of two sets, the set of all non-zero integers $\{0, 1, 2, \dots\}$ and the set of all negative integers $\{-1, -2, -3, \dots\}$. The enumerations given suggest how to show that each is countable.

Lemma 1.4.4. Any finite union of countable sets is countable.

Proof Hint: Use Lemma 1.4.3 and argue by mathematical induction on the number of unions.

Construction of $\mathbb{Z}$ from $\mathbb{N}$

Def. Recursive definition of addition on $\mathbb{N}$:

$$\begin{aligned}n + 0 &= n \\ n + S(m) &= S(n + m)\end{aligned}$$

Let $m, n, p, q \in \mathbb{N}$.

Def. $(m, n) \equiv_Z (p, q)$ iff $m + q = p + n$.

Example: $(2, 5) \equiv_Z (9, 12)$.

Def. $\mathbb{Z} =_{df} (\mathbb{N} \times \mathbb{N}) / \equiv_Z$.

Example: $-3 = [(0, 3)]_{\equiv_Z}$.

Def. $[(m, n)]_{\equiv_Z} + [(p, q)]_{\equiv_Z} = [(m + p, n + q)]_{\equiv_Z}$.

Countably Infinite Sets (cont.)

Lemma 1.4.5. The Cartesian product of countable sets is countable.

Proof. It suffices to show that $\mathbb{N} \times \mathbb{N} \preceq \mathbb{N}$. Consider the mapping $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ given by

$$f(n, m) = 2^{n+1} \cdot 3^{m+1}.$$

Since each natural number greater than 2 has a unique prime decomposition, f is 1-1. ■

Corollary. The set $\mathbb{Q}$ of rational numbers is countable.

Proof. By the lemma and the corollary to Lemma 1.4.3, $\mathbb{Z} \times \mathbb{Z}$ is countable. Each rational number has a unique representation as the ratio of two integers p/q with no common divisors. The mapping $f : \mathbb{Q} \rightarrow \mathbb{Z} \times \mathbb{Z}$ given by $f(p/q) = (p, q)$ is obviously 1-1. So, $\mathbb{Q} \preceq \mathbb{Z} \times \mathbb{Z} \preceq \mathbb{N}$. ■

The Set of All Finite Sequences on a Countable Set

A case of considerable importance for studying languages is the set of finite sequences over a countable vocabulary.

This is formally the same as determining the cardinality of the set of n -tuples, for arbitrary n , of elements from a countable set.

Lemma 1.4.6. If A is countable, then

$$A^{<\omega} =_{df} \{ \langle x_1, \dots, x_i, \dots, x_n \rangle \mid n \in \mathbb{N} \text{ and } x_i \in A \text{ for all } i \text{ s.t. } 1 \leq i \leq n \}$$

is countable.

The proof is left as an exercise.

The Set of All Infinite Sequences of 0's and 1's

Lemma 1.4.7. The set of all infinite sequences of 0s and 1s is uncountable.

Proof. Call this set $\mathcal{S}$. What we want to show is that $\mathcal{S} \not\preceq \mathbb{N}$.

We argue by *reductio ad absurdum*, i.e., by assuming, contrary to what we want to show, that $\mathcal{S} \preceq \mathbb{N}$ and then deriving a contradiction.

So let's assume that $\mathcal{S} \preceq \mathbb{N}$.

Since obviously $\mathbb{N} \preceq \mathcal{S}$, this is equivalent, by the Schröder-Bernstein theorem, to assuming that $\mathbb{N} \sim \mathcal{S}$; in other words, that there exists a bijection $s : \mathbb{N} \rightarrow \mathcal{S}$.

An Uncountable Set (cont.)

We choose 's' as a mnemonic aid. For, if we write s_n for $s(n)$, then the sequence $s_0, s_1, \dots$ allegedly enumerates all the members of $\mathcal{S}$.

Now list this infinite sequence of 0s and 1s running down the page. It will look something like this:

s_0 :	1	0	0	1	0	$\dots$
s_1 :	1	1	1	1	0	$\dots$
s_2 :	0	0	0	1	1	$\dots$
s_3 :	0	1	0	0	1	$\dots$
s_4 :	0	0	1	0	1	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Now scan down the diagonal of this infinite matrix starting from the upper left and running toward the lower right.

An Uncountable Set (cont.)

There you see the sequence beginning:

1 1 0 0 1 ...

If you “flip” each member of the sequence by replacing 1s with 0s and 0s with 1s, then you get the sequence beginning:

0 0 1 1 0 ...

This sequence cannot be any of the sequences listed going down. For it must disagree with each on the diagonal. Therefore, there is an infinite sequence of 0's and 1's that has been left off the list, contrary to the assumption that we had listed all of them.

N.B. See the supplemental slides for a rigorous formulation.

Corollary. $\mathcal{P}(\mathbb{N})$ is uncountable.

The proof is left as an exercise.

More Uncountable Sets

A similar kind of ‘diagonalization’ argument establishes the following:

Lem 1.4.8. The open set $\mathbb{R}(0, 1)$ of real numbers between 0 and 1 is uncountable.

Corollary. The set $\mathbb{R}$ of all real numbers is uncountable.

Lem 1.4.9. $\mathbb{R} \sim \mathbb{R}(0, 1)$.

The proof is left as an exercise.

$$\begin{aligned}
f(1) &= \{ 1, \dots \} \\
f(2) &= \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, \dots \} \\
f(3) &= \{ 2, 3, 4, 6, 8, 10, \dots \} \\
f(4) &= \{ 1, 3, 4, 5, 7, 9, 11, \dots \} \\
f(5) &= \{ 1, 2, 4, 5, 6, 7, 9, 11, \dots \} \\
f(6) &= \{ 3, 4, 6, 7, 9, 10, \dots \} \\
f(7) &= \{ 1, 5, 7, 9, \dots \} \\
f(8) &= \{ 3, 4, 7, 8, 11, \dots \} \\
f(9) &= \{ 1, 2, 5, 6, 9, 10, \dots \} \\
f(10) &= \{ 1, 2, 4, 5, 6, 9, 10, 11, \dots \} \\
f(11) &= \{ 1, 2, 4, 6, 9, 11, \dots \} \\
&\vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \vdots \quad \ddots
\end{aligned}$$

$$T = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, \dots \}$$

$\mathbb{N}$ does not have the same cardinality as its power set $\mathcal{P}(\mathbb{N})$: For every function f from $\mathbb{N}$ to $\mathcal{P}(\mathbb{N})$, the set $T = \{n \in \mathbb{N} : n \notin f(n)\}$ disagrees with every set in the range of f , hence f cannot be surjective. The picture shows an example f and the corresponding T ; **red**: $n \notin T$, **blue**: $n \in T$.

Cantor's Theorem



Figure: Georg Cantor (1845–1918)

Cantor's Theorem

We have seen that $\mathbb{N} \prec \mathcal{P}(\mathbb{N})$. Here is a more general result:

Theorem 1.4.1. For any set X , $X \prec \mathcal{P}(X)$.

Proof. That $X \preceq \mathcal{P}(X)$ is pretty obvious: just map each $a \in X$ to $\{a\}$.

Slightly harder is to show that $\mathcal{P}(X) \not\preceq X$. We argue again by *reductio ad absurdum*.

Suppose that $\mathcal{P}(X) \preceq X$. Then by the Schröder-Bernstein Theorem, $X \sim \mathcal{P}(X)$, and hence, there is a bijection $f : X \rightarrow \mathcal{P}(X)$. We show this is impossible. Let

$$Y =_{df} \{a \in X \mid a \notin f(a)\}.$$

Since $Y \in \mathcal{P}(X)$ and f by hypothesis is surjective, there exists a $b \in X$ s.t. $Y = f(b)$. Now ask whether or not $b \in Y$.

Cantor's Theorem (cont.)

For sake of reference, we redisplay the set Y .

$$Y =_{df} \{a \in X \mid a \notin f(a)\}.$$

Suppose that $b \in Y$. Then, by the definition of Y , $b \notin f(b)$.

But, by hypothesis, $f(b) = Y$. So, it follows that $b \notin Y$.

Alternatively, suppose that $b \notin Y$. Then, by the definition of Y , $b \in f(b)$.

But, again, since by hypothesis $f(b) = Y$, it follows that $b \in Y$.

So, $b \in Y$ iff $b \notin Y$, which is a classical inconsistency. ■

Endlessly Large Cardinalities

Since for *any* set X , $X \prec \mathcal{P}(X)$, we can continue to iterate the power set operator on $\mathbb{N}$:

$$\mathbb{N} \prec \mathcal{P}(\mathbb{N}) \prec \mathcal{P}(\mathcal{P}(\mathbb{N})) \prec \mathcal{P}(\mathcal{P}(\mathcal{P}(\mathbb{N}))) \prec \dots .$$

Let

$$\begin{aligned}\mathcal{P}^0(X) &=_{df} X \\ \mathcal{P}^{n+1}(X) &=_{df} \mathcal{P}(\mathcal{P}^n(X)).\end{aligned}$$

Then, if $\{\mathcal{P}^n(\mathbb{N}) \mid n \in \mathbb{N}\}$ is a set,

$$\bigcup \{\mathcal{P}^n(\mathbb{N}) \mid n \in \mathbb{N}\}$$

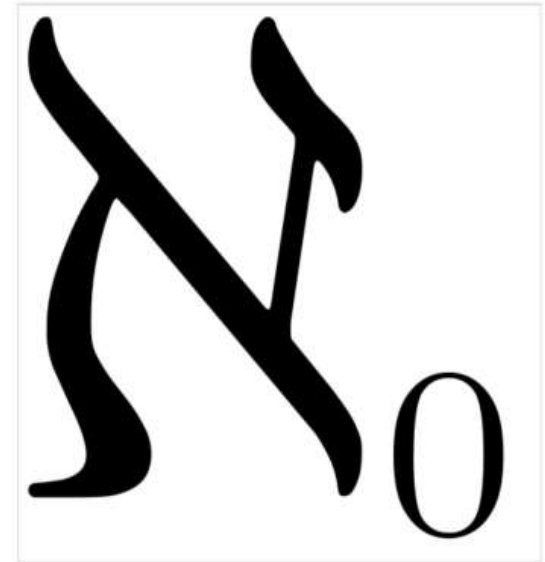
gives us an even bigger set. Cantor was so smart, he showed that for any uncountable cardinality, we can get that many more even larger cardinalities. David Hilbert referred to this as *Cantor's Paradise*.

Aleph numbers

The aleph numbers are a sequence of cardinal numbers that represent the sizes of infinite sets, denoted with an aleph $\aleph$, the first letter of the Hebrew alphabet.^[75] The first aleph number is $\aleph_0$, called "aleph-nought", "aleph-zero", or "aleph-null", which represents the cardinality of the set of all natural numbers: $\aleph_0 = |\mathbb{N}| = |\{0, 1, 2, 3, \dots\}|$. Then, $\aleph_1$ represents the next largest cardinality, then $\aleph_2$, and so on.^[76] The most common way this is formalized in set theory is through Von Neumann ordinals, known as Von Neumann cardinal assignment.^[77]

Ordinal numbers generalize the notion of *order* to infinite sets. For example, 2 comes after 1, denoted $1 < 2$, and 3 comes after both, denoted $1 < 2 < 3$. Then, one defines a new number, ω , which comes after every natural number, denoted $1 < 2 < 3 < \dots < \omega$. Further $\omega < \omega + 1$, and so on.^[78] More formally, these ordinal numbers can be defined as follows:

$0 := \{\}$, the empty set, $1 := \{0\}$, $2 := \{0, 1\}$, $3 := \{0, 1, 2\}$, and so on. Then one can define $m < n$, if $m \in n$, for example, $2 \in \{0, 1, 2\} = 3$, therefore $2 < 3$. Defining $\omega := \{0, 1, 2, 3, \dots\}$ gives ω the desired property of being the smallest ordinal greater than all finite ordinal numbers. Further, $\omega + 1 := \{1, 2, \dots, \omega\}$, and so on.^[79]



Aleph-nought, aleph-zero, or aleph-null: the smallest infinite cardinal number, and the cardinal number of the set of natural numbers.^[75]



Figure: David Hilbert (1862-1943)

The Power of the Continuum

So, we have seen that $\mathbb{N}$ is the smallest infinite set in cardinality.

And we have seen that $\mathbb{N} \prec \mathcal{P}(\mathbb{N}) \sim \mathbb{R}$.

Is there a set A such that $\mathbb{N} \prec A \prec \mathbb{R}$? Cantor couldn't find one, so he conjectured:

Cantor's Continuum Hypothesis (CH). There is no set A such that $\mathbb{N} \prec A \prec \mathbb{R}$.

But Cantor could not prove **CH**. Nor could Gödel. Nor could Gödel prove $\neg\mathbf{CH}$.

But Gödel could show

- if **Z** is consistent, then **Z + AC** is consistent,
- and if **Z + AC** is consistent, then **Z + AC + CH** is consistent.



Figure: Kurt Gödel (1906-1978)